

Kweku Andoh Yamoah

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Education

University of Florida, Ph.D. Computer Science Aug 2025 – Present

- CGPA: 4.0/4.0

Ashesi University, BSc. Computer Science Sep 2018 - May 2022

- CGPA: 3.95/4.00 (Top 1%)
* Graduated with *Summa Cum Laude*, with highest academic honors.

Research Interest

Neuro-Symbolic AI; Multi-Agent Systems; Multilingual Low-Resource NLP (African Languages); Efficient and Controllable Language Generation.

I investigate neuro-symbolic and multi-agent approaches to NLP in low-resource contexts. My goal is to develop systems that utilize structured linguistic knowledge to facilitate efficient, reliable, and controllable language understanding and generation, with a focus on African languages.

Publications

An Elicitation-Matrix Approach to Pragmatic Context Modeling in Low-Resource Machine Translation: The Case of Akuapem Twi May 2026

Kweku Andoh Yamoah, Godfred Agyapong, Kevin C. Scroggins, Neel Parekh, Chathuri Jayaweera, Shlok Gilda, Bonnie Dorr, Emmanuel J. Dorley

In *The International FLAIRS Conference Proceedings*, 39(1), doi: 10.32473/flairs.39.1.141846

An Elicitation-Matrix Approach to Pragmatic Context Modeling in Low-Resource Machine Translation: The Case of Akuapem Twi May 2026

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Winning Isn't Reasoning: Evaluating Iterative Reasoning Updating in Language Models May 2026

Kevin C. Scroggins, **Kweku Andoh Yamoah**, Emmanuel J. Dorley

In *The International FLAIRS Conference Proceedings*, 39(1), doi: 10.32473/flairs.39.1.141832

Enhancing Speech instruction Understanding and Disambiguation in Robotics via Speech Prosody Aug 2025

David Sasu, **Kweku Andoh Yamoah**, Benedict Quartey, Natalie Schluter

In *Proc. Interspeech 2025*, doi: 10.21437/Interspeech.2025-1961

Building a Functional Machine Translation Corpus for Kpelle July 2025

Kweku Andoh Yamoah, Jackson Weakoo, Emmanuel J. Dorley

In *Proceedings of the Sixth Workshop on African Natural Language Processing (AfricaNLP 2025)*, Association for Computational Linguistics, doi: 10.18653/v1/2025.africanlp-1.8

Evaluating Deepfake Speech and ASV Systems on African Accents Sep 2023

Kweku Andoh Yamoah, Hussein Baba Fuseini, David Ebo Adjepon-Yamoah, Dennis Asamoah Owusu

In *Deep Learning Indaba 2023*

Under Review

Fine-Tuning A Large Language Model for Systematic Review Screening

Kweku Andoh Yamoah, Noah Schroeder, Emmanuel Dorley, Neha Rani, Caleb Schutz

Under review at: 27th International Conference on Artificial Intelligence in Education, *Main Conference*.

Context-Switching Costs in Hybrid LLM-Algorithm Systems

Kevin C. Scroggins, **Kweku Andoh Yamoah**, Kevin Kurian, Emmanuel J. Dorley

Under review at: AAMAS 2026 Workshop on Strategic Engineering

AI Assisted Data Extraction for Systematic Reviews in Education

Noah Schroeder, Chris Davis Jaldi, Shan Zhang, **Kweku Yamoah**

Under review at: Educational Technology Research and Development

Professional Experience

Graduate Research Assistant, University of Florida, CISE – Gainesville, FL Aug 2025 – Present

Graduate Research Assistant in the Intelligent Agents Research Lab under Prof. Emmanuel Dorley, focusing on multilingual low-resource NLP for African languages. Responsibilities include building language datasets, developing neuro-symbolic representations, fine-tuning language models for tasks such as translation and speech, and documenting reproducible experimental pipelines to produce publication-ready results.

Undergraduate Teaching Assistant, Ashesi University – Berekuso, Ghana Aug 2022 – March 2025

Courses handled: Machine Learning, Introduction to Artificial Intelligence, Natural Language Processing, Discrete Structures & Theory, and Networks & Data Communications, Cloud Computing, Intermediate Computer Programming in C++.

Awards

Best Paper Finalist in Datasets Track DeepLearning Indaba, 2025 August 2025

Best Computer Science Senior Capstone Project, Ashesi University June 2022

Dean's List, Ashesi University 2019, 2020, 2021, 2022

MasterCard Foundation Scholar, Ashesi University 2019, 2020, 2021, 2022

Invited Talk

Natural Language Processing: An Overview, *Introduction to Artificial Intelligence (CAP 4621)*, University of Florida, Gainesville, FL Nov 2025

Academic and Professional Services

Reviewer

Feb 2026 2026 African NLP Workshop

University Service

Academic Committee Co-Chair, Ashesi Student Council Feb 2021 - May 2021

Faculty Intern for Ashesi Success, Ashesi University Feb 2021 - May 2021

Coach for the Coaching, Counseling, & Academic Peer Support Group, Ashesi University Sep 2019 - Nov 2021

Volunteer Service

Hackathon Judge, Ashesi Design Lab 2022, 2023

ML Session Tutorial Lead, Maker's Session, Ashesi Design Lab 2023

Moderator, MasterCard Foundation Scholars Convening 2021

Technologies

Main Programming: Python

Familiar Programming Language: C++, C, Java, SQL

ML libraries: PyTorch, TensorFlow